



DARTMOUTH
ENGINEERING
THAYER SCHOOL

ENGS 75 Final Report

Team 5: Port-A-Dock

Introduction & Background:

With three team members having a background in rowing, our group was familiar with the struggles associated with launching and docking small shells. Whether it is the risk of flipping when entering or exiting the shell or your shell floating away while you are getting ready to go out, there is a problem to be solved. To further understand what issues exist surrounding rowing shells and the dock, our group conducted interviews with several individuals. The individuals had varying rowing experience and ranged from elite rowers to novices. We focused our research on the master's rowing community interviewing several members of the Upper Valley Rowing Foundation (UVRF). We quickly found that there were two common issues when it comes to rowing small shells: a lack of stability and the risk of one's shell floating away. Quotes from the interviews highlight the extent of the problems as individuals interviewed stated: "Once you get upwards of 70, you tend to have to crawl out of the

shell”, “I am constantly worried about my single floating away”, and “The risk of flipping restricts my ability to row the single.” After conducting the interview and organizing the results, we then prioritized the needs.

Need-finding Requirements:

Looking at the results of the interviews, the group was able to identify three main needs. The first was stability, all interviewees cited fear of capsizing on the dock and how they would love a device that could provide a sense of security. Secondly, the group decided to focus on creating a device that would provide a fastening capability. Boats left unattended on moving bodies of water often float away if left unattended and could sustain extensive damage. Finally, we decided to prioritize ease-of-use. Given that our target market mainly consists of masters and novice rowers, it is critical that our product is simple to use. Especially when considering the instability involved in launching and docking, it is important to make sure that using our product does not further complicate those processes. These three needs served as the pillars of our value proposition: A device that will stabilize and secure small shells, providing ease of mind for users when launching and docking and improving ease of use for everyone in the sport, and making it more friendly for beginners.

Specifications:

After identifying the three main needs and creating our value proposition. The group then created a set of specifications we wanted our device to meet. Again, we first focused on stability. Our device would ideally keep all parts of the boat besides the hull out of the water for at least 10 minutes. As well, the device would prevent capsizing when 100 lbs is applied to the side away

from the dock and have the ability to secure both stern and bow-mounted outrigger style boats. Next, we looked at what we wanted the fastening capability of our device to entail. We decided that we wanted our device to be able to secure the single to the dock for at least 10 minutes in moving water. In addition, we considered environmental factors when considering the fastening capability of our device and concluded that we wanted our device to keep the boat tethered to the dock in 10,000 CFS and in winds up to 10 mph. We also determined that the device must not obstruct or interfere with current use of the dock or risk damaging boats. Finally, when considering ease of use, we decided that our device should be able to be latched and unconnected within 10 seconds, as well as that the device should be operable with one hand.

Tests & Acceptance Criteria:

The next step after determining the specifications was creating tests to determine if our device met the desired criteria. Each specification and its corresponding test, as well as the criteria needed to be met in order to qualify as a pass, are highlighted below.

Specification	Test	Criteria
Prevents capsizing when 100 lbs is applied to the side away from the dock	Apply 100lbs to the opposite edge of hull; Measure the angle change of the hull from horizontal	Pass: hull angle change is <20°
Can secure both stern and bow-mounted rigger boats	Will run all validation tests on both stern and bow-mounted shells.	Pass: both iterations of shells pass all tests

Must be able to secure the single to the dock for at least 10 minutes in moving water Can keep the boat tethered to the dock in 10,000 CFS Secures the boat to the dock in winds up to 10 mph Keeps all parts of the boat besides the hull out of the water for at least 10 minutes	Single shell for 10 minutes at the dock in maximum conditions: 10,000 cfs and 10 mph winds, test in 3 different wind directions.	Pass: Shell stays at the dock Pass: zero hits of hull on dock
Can support 200 lbs of weight	200 lbs force applied horizontally to the farthest point from the dock	Pass: No permanent deformation
Does not interfere with the docking/launching processes regardless of whether the product is in use	Observe 3 launches before installation, after installation using the product, and not using the product	Pass: No change in launch time or difficulty
Can be utilized with one hand	Observe 3 uses with two hands and 3 uses with two hands	Pass: No change in ease of use

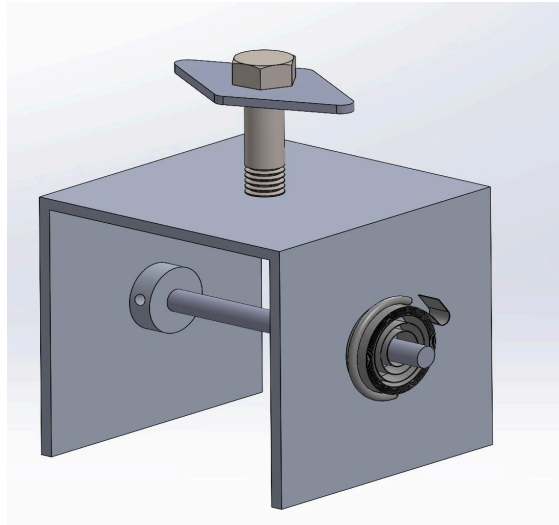
Design:

With the tests and criteria determined, we began to brainstorm ideas. We began by thinking broadly, focusing on different aspects of the design, such as integration, life cycles, additional features, and even the product as a service. We spent substantial time considering whether the device would be mounted to the dock, the shell or both. Additionally, we considered what materials would endure the elements the best, but how to design the device to allow for easy repair in the case of degradation. Some of the additional features we considered adding included a display for conditions such as temperature and wind speed. As well, we contemplated integrating software into the device, which would allow one to track how many people were on

the water at a given time. Afterwards, we generated myriad designs and used a selection matrix to select one concept to move forward with. The designs included a spring-loaded hook for the rigger, a folded gate latch, a bungee cord, a shell clamp, and a ramp for the hull. We then utilized the concept selection matrix displayed below to decide which concept to move forward with

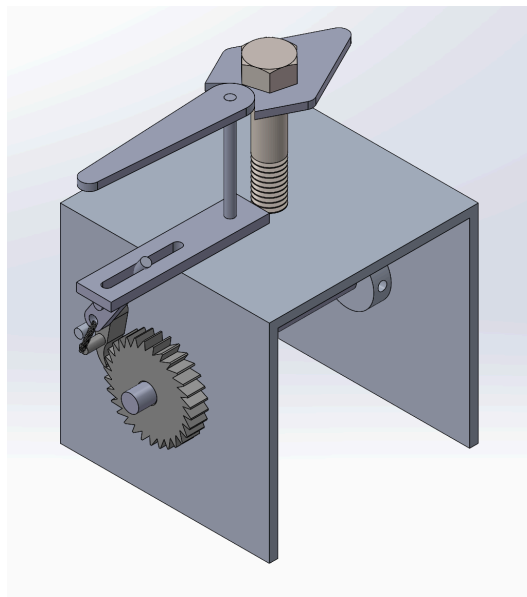
Concept	Stability/Tether (2x)	Obstruction (2x)	Ease of use (1x)	Total
Retractable Strap	6	10	3	19
Rigid Retraction	8	8	2	18
Hull Mount	10	2	4	18

After completing the selection matrix, we decided to move forward with the retractable strap. A seatbelt-esque device that exists underneath the dock and uses a seatbelt-like mechanism to secure the shell to the dock. It wraps around the rigger and clips back to itself, locking the boat in place. Once we had narrowed down the final concept, we began to model the device in SolidWorks, creating several iterations throughout the process. The first model was a design inspired by a seatbelt. This design lacked a locking mechanism but utilizes a rotary spring, includes ball bearings, and is designed to fit in the opening in the dock.



Design 1

The second model improved upon the first design. The key update is a ratchet and pawl mechanism, which would secure the gear in place upon locking preventing the strap from extending. This provided a user-friendly method to lock the rotary spring, but the ratchet did not fit within the dock opening, and required multiple actions when securing the boat.



Design 2

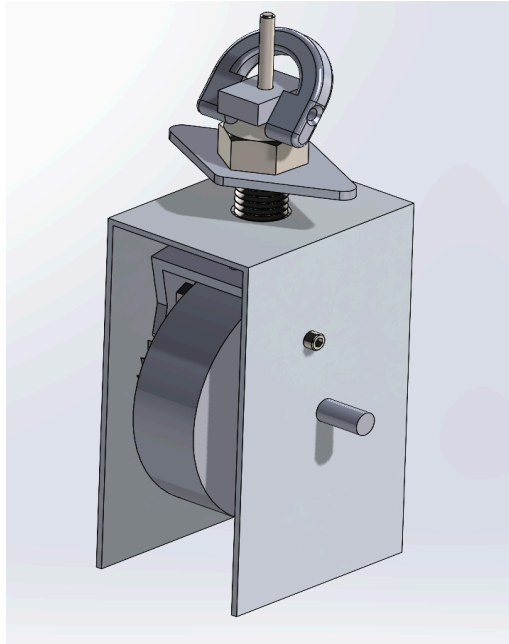
When designing the next iteration, we sought to create a ratchet and pawl mechanism that would fit within the dock opening. The design utilizes a slot in the housing to ensure that the switch that

actuates the pawl fit does not protrude from the device. This design also moved the ratchet gear and rotor spring internally to the housing. Another benefit of this design is that the slot that switch travels within is easier to manufacture and requires fewer parts, however it does not reduce the number of steps required to lock or unlock the boat.



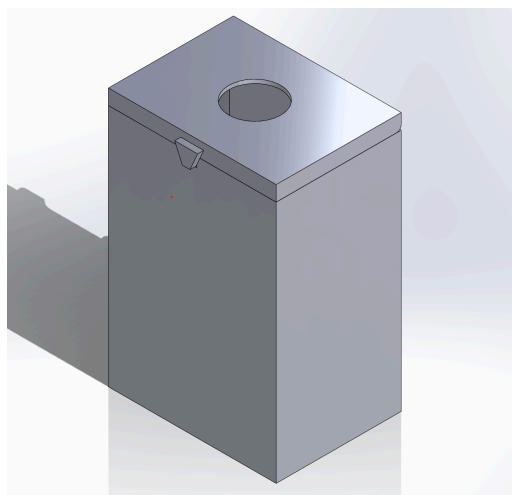
Design 3

The fourth iteration featured what ended up being a big design breakthrough. The design incorporated a cam mechanism actuated by the loop that you hook back to, to engage/lock the ratchet and pawl. This made the device more user-friendly by reducing steps for lock and unlock. Now inherently when strap is hooked back to the dock, the ratchet is locked so the the strap cannot extend. There were many edits made within this design to integrate the rotor spring and constrain the center shaft while allowing the hub and outside of the rotor spring to rotate.



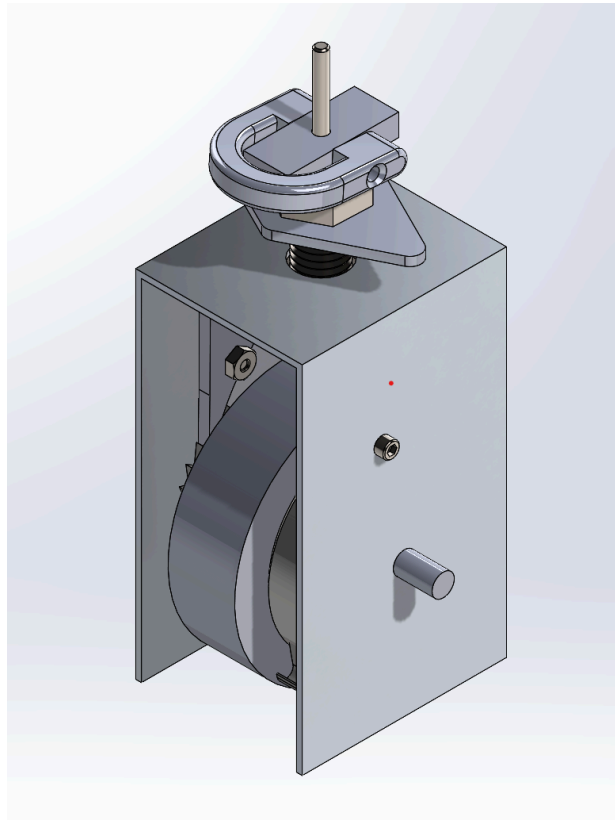
Design 4

While the final design might not be considered a full design, we decided to model a protective shell in order to prevent the device from degrading in the elements. This would protect internals of the device from the sun and things that could hit or get stuck in the moving components. The device would utilize a snap-fit mechanism. This shell was never prototyped due to time constraints, but would be in future iterations.



“Design” 5

After several iterations, the group settled on a final design. It utilizes the cam mechanism to engage and disengage the ratchet and pawl. Fits within the dock opening, is user-friendly, and can be mass-produced with moderate ease. The design is almost identical to the fourth iteration, however features a reworked rocker and pawl and is re-dimensioned to fit within the dock.



Final Design

The group applied several of the concepts taught in class when creating the various iterations. The group considered designing for manufacturing, attempting to minimize the cost of components and transportation by keeping the device small. This allowed for fewer total and custom parts which resulted in an easy assembly process that required few tools. Next the group considered designing for the environment. By making the product easily repairable, it allowed for a long life cycle. When also considering DFE, the group attempted to minimize the amount of raw materials used. Additionally, the group designed for robustness, the materials chosen

throughout the process were able to withstand the environment. As well, the device was designed to be easily utilized by an individual with little to no prior knowledge of its functionality. Finally, the group designed and iterated with a number of miscellaneous factors in mind. It was important to the group that the device could fit within a standard dock opening and be removed and installed with ease. It was also critical that the device could work with all models of rowing shells.

Prototyping:

The group also created prototypes in parallel with the design process. The first prototype consisted of the outer shell and spool and was the simplest iteration of the concept. This prototype mainly served as a way for the group to visualize the overall design and how it would integrate within the dock.



Prototype 1

Unfortunately, the group did not document our second prototype. However, it was an exact physical manifestation of Design 2 featured above. This was our First attempt at ratchet and pawl which worked quite well. Through some simple testing of the mechanism, we realized that it

required additional steps to lock/unlock which made it difficult to use and prone to error. For example, if the user hooked the sprap in, but forgot to flip the switch that engaged the pawl, the strap could continue to unroll and allow the boat to float from the dock. This allowed us to see how the mechanism could be simplified and integrated with the action of hooking the end of the strap back to the device.

The third prototype was the group's first attempt at the cam mechanism for engaging the pawl and consisted of mostly 3D printed parts. It used a loop and cam at the top that would pull a shaft up and down to rotate a rocker that moved the pawl in and out of the interface with the ratchet gear. This proved that the concept worked, but was not easy to manufacture with more robust materials such as aluminum. This prototype also did not capture the center of the rotor spring and prevent rotation, so there was to force to retract the strap.



Prototype 3

The fourth prototype was our final. The device is fully functional and serves as a good test device and proof of concept, but still need some improvement and tuning. The main shaft

consists of a bolt with a slot to hold the center of the rotor spring, and a press fit washer and pin constrain the shaft to the body and not allow rotation. The hub and roll of strap rests of 2 bearings allowing it to rotate about the shaft winding up the rotor spring. The device also features an improved pawl design matches the movement of the pawl with the thickness of the hook. Unfortunately the pawl was still made from 3D printed PLA, but is intended to be aluminum in the future. After the completion of this prototype the group was able to move on to testing.



Prototype 4

Testing:

As previously stated, the group hoped to assess the stabilizing, tethering, and ease-of-use capabilities of the device during testing. Unfortunately, the time window to test the prototype was extremely limited, and the group was not able to conduct as comprehensive a testing process as desired. However, the few tests conducted bore promising results. The prototype functioned as imagined: the rigger was securely fastened to the dock. The boat was not able to be pried off the dock with significant force applied, and the shell remained secure to the dock in moderate flow.

The device was easy to use and completed the functions that we intended from the beginning. It held the shell at the dock in normal conditions and prevented the hull from capsizing when unbalanced forces were applied. Due to time constraints we were unable to more specifically test limitation of the product including how much weight it could withstand, how much angle of rotation it would prevent, or how well it helped a shell in extreme conditions such as more flow or high winds. The testing also highlighted that the installation and removal process was extremely tedious. There were too many components to assemble at one time in a small and hard to access place.



Testing Setup

Documentation:

Our files were stored in the ThayerFS team folder, and we used ThayerFS to share ideas and work on our latest models. Here is a brief summary of what we have stored in our shared ThayerFS folder:

- Media from our market research
- CAD models of our four design revisions/iterations (Labeled Rev A-D)
- Drawings for any parts that were machined
- STLs for any items that were 3D printed

We generated two bills of materials in parallel—a bill of materials for our active prototype, as well as a presumptive bill of materials for a production model. These BOMs are displayed below:

Prototype Bill of Materials

Bill of Materials							
Product: Port a dock - prototype							
Team: 5							
Date: 6/6/2025							
Part ID #	Description	Ind. part or sub-assembly	Quantity	Function (verb-noun)	Material	Manufacturing method	Cost estimate (hi, med, low)
1	Housing		1	Holds mechanism	Aluminum	Sheet metal (punching, bending)	n/a (prototype)
2	Axle		1		Aluminum	Stock followed by groove cut and press fit	n/a (prototype)
3	2 inch 1/2-13 hex bolt		1	Fixes mechanism to dock	Steel	Purchased	n/a (prototype)
4	Hex nut 1/2-13		2	Receives bolt	Steel	Purchased	n/a (prototype)
5	Diamond washer		1	Maintains fit with dock	Steel	Machined	n/a (prototype)
6	Spool			Holds spring, webbing, gear	PLA	3d Print	n/a (prototype)
7	Nylon nut #4-40		3	Secures pall	Steel/Nylon	Purchased	n/a (prototype)
8	Ball bearing		2	Holds axle, allows rotation	Steel	Purchased	n/a (prototype)
9	Ratcheting gear		1	Enables locking	Steel	Purchased	n/a (prototype)
10	Cap screw #6-32 2.75"		1	Connects cam and pawl	Steel	Purchased	n/a (prototype)
11	Cap screw #4-40 5/16"		3	Holds pawl, prevents rotation	Steel	Purchased	n/a (prototype)
12	Cam axle housing		1	Holds cam	Aluminum	Machined	n/a (prototype)
13	Hook cam		1	Tightens pawl, attaches hook	Aluminum	Machined	n/a (prototype)
14	Rotor spring		1	Retracts strap	Steel	Purchased	n/a (prototype)
15	Pawl		1	Engages locking	PLA	3d Print	n/a (prototype)
16	Strap (string)		1	Holds rigger	Fabric	Purchased	n/a (prototype)
17	Hook	Subassembly	1	Attaches rigger	Steel	Purchased	n/a (prototype)
18	Fender washer		1	Prevents rotation of axle	Steel	Purchased	n/a (prototype)
19							

Production Model Bill of Materials

Bill of Materials								
Product: Port a dock - prototype								
Team: 5								
Date: 6/6/2025								
Part ID #	Description	Ind. part or sub-assembly	Quantity	Function (verb-noun)	Material	Manufacturing method	Cost estimate (hi, med, low)	Further notes
1	Housing		1	Holds mechanism	Aluminum	Sheet metal (punching, bending)	Med	
2	Axle		1		Aluminum	Stock followed by groove cut and press fit	Low	
3	2 inch 1/2-13 hex bolt		1	Fixes mechanism to dock	Steel	Purchase/Supplier	Low	
4	Hex nut 1/2-13		2	Receives bolt	Steel	Purchase/Supplier	Low	
5	Diamond washer		1	Maintains fit with dock	Steel	Stamping	Low	High upfront cost
6	Spool			Holds spring, webbing, gear	Aluminum	Machined (turning, milling)	High	
7	Nylon nut #4-40		3	Secures pawl	Steel/Nylon	Purchase/Supplier	Low	
8	Ball bearing		2	Holds axle, allows rotation	Steel	Purchase/Supplier	Med	
9	Ratcheting gear		1	Enables locking	Steel	Machined	High	
10	Cap screw #6-32 2.75"		1	Connects cam and pawl	Steel	Purchase/Supplier	Low	
11	Cap screw #4-40 5/16"		2	Holds pawl	Steel	Purchase/Supplier	Low	
12	Cam axle housing		1	Holds cam	Aluminum	Die casting	Med	
13	Hook cam		1	Tightens pawl, attaches hook	Aluminum	Die casting	Med	
14	Rotor spring		1	Retracts strap	Steel	Purchase/Supplier	High	
15	Pawl		1	Engages locking	Steel	Die casting	Med	Optimized for DFM - no overhangs!
16	Strap		1	Holds rigger	Rubberized fabric	Purchase/Supplier	Low	
17	Custom Hook	Subassembly	1	Attaches rigger	Steel	Machined	High	
18	Fender washer		1	Prevents rotation of axle	Steel	Stamping	Low	

The main differences between the prototype and production would be optimizing the production of the custom parts including materials and production technique. For example, the pawl would be diecast out of stainless steel, all of the machined parts would be CAM machined out of aluminum with programs to minimize labor and improve precision. The purchased components such as bolts, nuts, springs, and bearings would be purchased in bulk significantly reducing their individual costs. We would also spend more time finding the optimal components to purchase. For example the rotor spring we used in the prototype was much larger than needed, but there were few options available on McMaster Carr so we used a larger more expensive component that we would optimally want. For production we would be able to purchase from a wider range of suppliers and get a more optimal component at a better price.

We would also optimize the assembly process with jigs and custom assembly tools to maximize efficiency and precision. One example would be an alignment jig that holds the hub and housing concentric so that the shaft could be inserted with minimum difficulty. Another benefit of our design is that almost all components are stainless steel or aluminum so they would not degrade over time allowing us to provide a refurbishment service for our devices and replacement parts for the wear components such as bearings, pawl, and cam.

We do not have sufficient industry experience to provide a precise estimate of the theoretical shipped cost of our product. However, based on the limited number of high-cost items

and the fact that the model is largely optimized for DFM/DFA, it is likely that this product would be in the range of \$100-200 shipped in smaller-scale production. This is consistent with other products of similar complexity, material, and precision. While this may sound high for a small device in recreational application, the nature of the product's market makes and intended use makes the price reasonable. For one, the rowing community is generally well-off and not afraid to invest in the sport. For example, many shells cost upwards of tenthousand dollars. Furthermore, the product would be highly reliable and durable. We hope that it could be used for ten year or more with minimal maintenance. Maintenance would likely be replacing bearings and straps every few years which would cost less than twenty dollars and only take a few minutes. We envision this as something that a boathouse manager can install and forget about. It is intended to be installed at rowing organizations or team docks where a team would invest in outfitting their docks with a couple of our products such as one every 10ft of dock. Therefore every user would only be investing a small percentage to of the purchase cost. We believe that the product would provide extensive value to the rowing community without significant sticker shock.

Learning:

Although the project was largely successful, upon reflection, there is much the group would do differently to both improve the product and the process. First, the group would machine the pawl out of stainless steel to increase its strength and precision. This did not occur due to time constraints and resulted in the breaking of weaker 3D printed pawls. Secondly, the group would create a protective shell to protect against the elements and to guide the strap in and out of the device so that it rolls on the hub neatly. Again, this did not occur due to time

constraints. We would also increase mounting bolt size to reduce movement during installation as well as mount the bolt to the housing for an easier installation process. In regard to the process, the group should have began the prototyping and testing process earlier in the design process. Such as, testing and recording the problem itself at the dock. Testing some extremely simple prototypes such as a rope and snap tied to the dock. Then as the design developed we could have made some sub-assembly prototypes such as the hub, shaft, and spring, or the ratch pawl assembly and tested those individually. This would have allowed for faster iterations and more trial and error. We were also very limited by when our purchases components arrived partially because we waited until we knew exactly what components we needed to order, but we could have made a larger order earlier in the design process with options of components that we could test. This would only work for inexpensive components though.